

Musculoskeletal Digital Twin

Ontological Framework - Conceptual Foundation for a Lower-Body Digital Twin

Purpose	Define what exists in the system, what properties those entities possess, and how they relate before translating the model into software.
Initial scope	Lower-body sagittal-plane model centered on a simple squat, with a representative leg and head-arms-trunk segment.
Long-term role	Provide a stable knowledge structure for biomechanics, simulation, sensing, control, personalization, and clinically meaningful parameter changes.

Core Perspective

The ontology is the conceptual backbone of the digital twin. It separates the physical human, the mathematical model, the measurements used to update that model, the task and environment in which movement occurs, and the conditions that modify the system. This separation helps prevent the software architecture from becoming a collection of loosely connected functions.

A useful guiding chain is: Condition -> Parameter change -> Dynamics change -> Movement change -> Outcome change.

1. Physical System

Represents the real biological and mechanical system being modeled.

Entity / Concept	Brief ontological meaning
Human	The physical subject whose movement and properties are represented by the digital twin.
Anatomical Structure	A general category containing body segments, joints, muscles, tendons, and ligaments.
Segment	A body treated as a mechanical unit, such as the foot, shank, thigh, or head-arms-trunk (HAT). It carries properties such as length, mass, center of mass, and inertia.
Joint	A connection between segments that constrains and permits relative motion. The initial squat model emphasizes the ankle, knee, and hip.
Muscle / Tendon / Ligament	Force-producing or force-transmitting biological structures. These can be introduced progressively as the model becomes more physiologically detailed.

2. Kinematics

Describes configuration and motion without yet explaining the forces that cause that motion.

Entity / Concept	Brief ontological meaning
Generalized Coordinate (q)	A compact variable describing system configuration, such as an ankle, knee, or hip joint angle.
Generalized Velocity (q-dot)	The time rate of change of a generalized coordinate.
Generalized Acceleration (q-double-dot)	The time rate of change of generalized velocity.
Pose and Position	Spatial descriptions of segments, joints, and the center of mass derived from the generalized coordinates and body geometry.

3. Dynamics

Explains why motion changes by relating mass properties, forces, moments, contact, and acceleration.

Entity / Concept	Brief ontological meaning
Force	An interaction capable of changing motion. Relevant examples include gravity, ground-reaction force, muscle force, ligament force, and other external forces.
Moment / Torque	Rotational effect of a force. Joint moments can arise from muscles, external loads, gravity, and contact.
Equations of Motion	The mathematical relationship between configuration, velocity, acceleration, internal actuation, and external forces. A common form is $M(q)\ddot{q} + C(q,\dot{q}) + G(q) = \tau + J(q)^T F_{ext}$.

Entity / Concept	Brief ontological meaning
Mechanical State	The set of variables needed to describe the system at a particular instant, such as joint position and velocity, with additional variables added as model complexity grows.

4. Neuromuscular and Control System

Represents how movement commands ultimately become mechanical forces and motion.

Entity / Concept	Brief ontological meaning
Controller	A rule, algorithm, or physiological abstraction that determines commands required to accomplish a movement task.
Neural Excitation	The command sent toward a muscle model.
Muscle Activation	A dynamic physiological state translating excitation into the muscle's force-producing capacity.
Muscle Force	Mechanical output of a muscle, influenced by activation and muscle-tendon properties.
Moment Arm	Geometric relationship converting muscle force into a joint moment.

5. Task and Environment

Separates the person from what the person is trying to accomplish and the external world in which it occurs.

Entity / Concept	Brief ontological meaning
Environment	Contains gravity, the ground, contact surfaces, external objects, and external loads.
Task	Defines a movement goal, initial conditions, constraints, and success criteria.
Squat	The initial task instance: begin from standing, descend, reach a lower configuration, and ascend while maintaining appropriate foot-ground interaction.
Constraint	A restriction on permissible motion or interaction, such as maintained foot-ground contact.

6. Digital Model and Twin

Distinguishes the physical human from the computational representation used for simulation and prediction.

Entity / Concept	Brief ontological meaning
Digital Human Model	A computational representation of the person's anatomy, mechanics, physiology, and current state.
Parameter	A relatively persistent characteristic of the model, such as segment length, mass, inertia, muscle strength, stiffness, or joint range of motion.

Entity / Concept	Brief ontological meaning
State	A quantity that evolves during movement, such as joint angle, joint velocity, muscle activation, or contact state.
Simulation	The process that evolves model state according to equations, parameters, inputs, constraints, and environmental interactions.
Digital Twin	The broader system formed when a digital model is connected to observations of a physical subject and can be updated, compared, or personalized using those observations.

7. Measurement and State Estimation

Provides the bridge between a physical person and the computational model.

Entity / Concept	Brief ontological meaning
Sensor	A source of observations, such as motion capture, inertial sensing, force measurement, or electromyography.
Measurement	An observed quantity, including kinematic, kinetic, physiological, or anthropometric data.
State Estimation	The process of using measurements to infer model states that may not be measured directly.
Parameter Estimation / Personalization	The process of adjusting model parameters so the digital representation better reflects a particular subject.

8. Condition and Clinical Modification

Represents impairments or clinically meaningful conditions as changes to defined model properties rather than as entirely separate models.

Entity / Concept	Brief ontological meaning
Condition	A named circumstance that modifies one or more parameters, constraints, or physiological properties.
Parameter Modification	The formal link between a condition and the computational quantity it changes.
Example: Quadriceps Weakness	Can be represented as reduced maximum force capacity in the relevant muscle model.
Example: Ankle Mobility Restriction	Can be represented as a change in allowable ankle range of motion or related stiffness properties.
Example: Tendon Stiffness Change	Can be represented by modifying the tendon stiffness parameter and observing downstream effects on force and movement.

9. Outcomes

Captures quantities used to interpret performance, mechanics, and potential clinical relevance.

Entity / Concept	Brief ontological meaning
Joint Loading	Forces or moments associated with a joint during the task.
Muscle Force	Estimated force generated by individual muscles or muscle groups.
Center of Mass	Whole-body or segment-level mass distribution used to characterize movement and balance.
Stability	Measures describing the relationship between body state, support, and the ability to maintain or recover desired motion.
Task Performance	Measures describing how successfully the movement goal is achieved.

10. Core Relationships

Source	Relationship	Target
Human	HAS	Segment / Joint / Muscle
Segment	CONNECTED BY	Joint
Muscle	CROSSES	Joint
Joint	HAS	Generalized Coordinate
Generalized Coordinate	DETERMINES	Segment Pose
Muscle Force	ACTS THROUGH	Moment Arm
Moment Arm + Muscle Force	PRODUCE	Joint Moment
Task	CONSTRAINS	Human / Digital Model
Human	INTERACTS WITH	Environment
Sensor	PRODUCES	Measurement
Measurement	UPDATES / ESTIMATES	State or Parameter
Physical Human	REPRESENTED BY	Digital Human Model
Condition	MODIFIES	Parameter
Simulation	EVOLVES	State
State + Parameters + Task	PRODUCE	Outcomes

11. Digital Twin Information Loop

Physical Human -> Sensors -> Measurements -> State / Parameter Estimation -> Digital Twin -> Simulation / Prediction -> Comparison with the Physical Human -> Model Update.

This loop distinguishes a digital twin from a static biomechanical model. The model is not only capable of simulation; it has an explicit conceptual pathway through which observations of the real system can inform and personalize the computational representation.

12. Translation into Software

The ontology should guide the codebase rather than merely document it. Ontological entities can become software abstractions, while ontological relationships can become interfaces and dependencies.

Ontology	Potential software representation
Human	HumanModel

Ontology	Potential software representation
Segment	Segment
Joint	Joint
Muscle	Muscle
Parameter	HumanParameters
State	HumanState
Environment	Environment
Task	Task
Squat	SquatTask
Condition	Condition
Measurement	Measurement
Simulation	Simulator

Recommended Next Step: Ontology v0.1

The next formal artifact should define only the entities required for the first sagittal squat model. Each entity should receive four fields: **definition, attributes, relationships, and computational representation**. This produces a controlled vocabulary that can directly guide repository structure and later expansion into muscles, sensing, personalization, and clinical conditions.

The ontology should remain extensible: the first implementation can be intentionally simple without changing the conceptual meaning of the entities when the model becomes more anatomically or physiologically sophisticated.